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Application of an unsteady three-dimensional computational fluid dynamics model to predictive habitat modelling of macroinvertebrates—A case study at the Ziller River in Austria

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Abstract

The Ziller River, a tributary to the Inn in Tyrol, Austria, is affected by hydropower generation. Macroinvertebrate habitat-preferences were investigated in the field and corresponding habitat suitability curves were applied successfully to a novel integrative assessment method based on hydrodynamic modelling. Here, a reach of the Ziller River is simulated using a three-dimensional unsteady hydrodynamic model. The three-dimensional simulation results are applied to a predictive macroinvertebrate habitat model. Habitat suitability indices (HSI) are calculated based on temporal alterations of bed shear stress values. Model results are free from two-dimensional flow-averaging approximations and demonstrate the critical information lost in lower-order models due to simplification procedures. Shear stress values, three-dimensional simulated velocity components, and temporal alterations of flow are provided in the current study. Considering the simulation results, the implementation of the provided supplementary simulation details in macroinvertebrate studies and the integration into further habitat investigations is discussed.

KEYWORDS

3D, CFD, habitat modelling, hydropowering, macroinvertebrates, unsteady

1 | INTRODUCTION

Fluctuations in river flow can have natural or anthropogenic causes (Barriendos & Rodrigo, 2006; Brázdil et al., 2011). Hydropower plants (HPP) are valuable resources that initiate anthropogenic causes related to on-demand energy production using storage HPPs (Dranka, Ferreira, & Vaz, 2020; Hauer, Holzapfel, Leitner, & Graf, 2017). In case of large fluctuations in energy demand, this rapid increase or decrease in the release of water from storage HPPs is inevitable. The rapid flow alteration is coined as hydropowering in literature (Bovee, 1985; Valentin, Lauters, Sabaton, Breil, & Souchon, 1996); hydropowering alters the flow regime in the downstream sections of HPPs. In contrast to Run-

of River HPPs, the operation of storage power plants causes artificial changes in flow on a daily and weekly basis, that is, peak and off-peak flow, which address high and low flow respectively. These frequent and regular events are totally different from natural flooding events and may affect the river ecology drastically (Charmasson & Zinke, 2011; Greimel et al., 2018). The dual nature of hydropowering (peak and off-peak condition) forms two hydrologically distinct “rivers”, in which organisms must be able to resist the abiotic variability (Jones, 2014). Abiotic impacts of hydropowering on rivers, that is, changes in hydrology, temperature, and sediment distribution, initiate the related impacts on aquatic organisms like benthic macroinvertebrates, fishes and macrophytes; this sequence addresses the

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cascading impacts of abiotic parameters and their influences on biotic indices. Moreover, the relation among abiotic and biotic parameters demonstrates comprehensive and complex interrelations in a river system and shows how small changes can affect ecologically sensitive habitats like streams and rivers (Person, 2013).

Benthic macroinvertebrates are excellent indicators of the ecological conditions in aquatic systems as a multitude of species have distinct habitat requirements. Moreover, the physical environment and disturbances due to hydropeaking on benthic populations are well-studied (e.g., Leitner, Hauer, & Graf, 2017; Schülting, Feld, & Graf, 2016). The alterations include shifts in longitudinal zonation, loss, and abundance reduction of specific taxa (Aldridge, Fayle, & Jackson, 2007; Lagarrigue et al., 2002), as well as impaired larval development (Frutiger, 2004). Drift or dislocation of macroinvertebrates can be classified as active (behavioural) or passive (accidental); the behavioural drift can be caused by sudden temperature alterations, high population density and competition etc. In contrast, catastrophic passive drift is often caused by an increase in water velocity and corresponding shear stress at the riverbed, which flushes benthic organisms or fish downstream (Céréghino et al., 2004; Carolli, Bruno, Siviglia, & Maiolini, 2012).

In riverine systems, the natural flow regime is an important ecological trigger for ecological integrity as well as species diversity (Dudgeon et al., 2006). The physical habitat of a river is affected by the corresponding flow regime and a mosaic of habitat features is generated and maintained by hydrologic variability (Poff et al., 1997). The variability of hydraulic habitats supports species diversity which have evolved according to system dynamic and discrete habitat conditions. Providing environmental flow (eFlow) in water management practices is essential to maintain variability of habitats and ecosystem services (Acreman et al., 2009). The determination of fluvial regime in rivers under hydropeaking can be addressed by developing various techniques in hydraulic habitat modelling (Hauer et al., 2017; Premstaller et al., 2016). Fundamental knowledge on interlinkages and processes of abiotic/biotic interaction, however, is frequently lacking. Experimental investigations are necessary to enhance basic understanding (Crowder & Diplas, 2002; Gualtieri, Abdi, Ianniruberto, Filizola, & Endreny, 2020; Piccolo, Frank, & Hayes, 2014), however, they are time consuming and labour intensive. Despite several studies, field or laboratory measurements are often limited by unavailability of expertise, appropriate equipment and risked by harsh weather and terrain conditions and cannot uncritically transferred to natural conditions.

Numerous efforts have been conducted to the development of numerical models as complimentary tools to facilitate habitat investigations (Benjankar, Tonina, & McKean, 2015; Guay et al., 2000; Holzapfel, Leitner, Habersack, Graf, & Hauer, 2017; Olivetti, Gil, Sridharan, & Hein, 2021). An example is the development of the Physical Habitat Simulation Method (PHABSIM) (Bovee, Lamb, Bartholow, Stalnaker, & Taylor, 1998), which has shown reliable results and good agreement between predicted and observed habitat parameters (e.g., Nagaya et al., 2008) for hydropeaking. The importance of three-dimensional flow investigations in hydraulics is already known and

discussed (Farhadi et al., 2018; Rütger & Olsen, 2007); especially, the crucial role of three-dimensional near-wall processes in reliability of numerical simulations is emphasised (Farhadi, Mayrhofer, Tritthart, Glas, & Habersack, 2018). However, only few modelling efforts, for example, (Shen & Diplas, 2010), were made to quantify the local and complex flow structures around obstructions and the associated aquatic habitats especially under hydropeaking conditions (Lamouroux, Hauer, Stewardson, & Poff, 2017). Moreover, detailed unsteady three-dimensional simulations are lacking so far. The issues are particularly even more important in Alpine fluvial systems, where the quality and abundance of habitats are influenced by the river coarse substrate. An example of such fluvial and habitat complications can be represented by reaches of the Ziller River in Austria, which are extensively affected by hydropeaking.

Assessment of the effects of hydropeaking in fluvial habitats is usually based on the instream flow incremental methodology (Bovee & O'Brien, 1982). This method relies on a quasi-steady flow approach coupled with a habitat suitability Index (HSI) (LeClerc et al., 1998; Booker et al., 2004; Holzapfel et al., 2017). The quasi-steady flow approach assumes an unsteady flow hydrograph as a sequence of constant discharges, that is, steady state flow, each of which proceeds for a certain period. The steady flow conditions, however, cannot precisely define the behavioural and morphological impacts of instantaneous energy slope alterations in hydropeaking. Therefore, to understand the effects of unsteady three-dimensional flow process on ecology of streams, it is necessary to simulate the currently ignored transition process from low to high flows during hydropeaking operations (Shen & Diplas, 2010).

Thus, this paper is presenting results of a novel integrative analysis on (i) “the habitat use” of selected macroinvertebrates and (ii) possible causes of absence of these macroinvertebrates based on “unsteady three-dimensional numerical simulations” results. Finally, we discuss the performance of such modelling techniques and their applicability to the current challenges facing habitat simulations. The work is focused on macroinvertebrates and their interactions with three-dimensional near-bed flow processes. Perception of such processes is essential for investigating the impact of flow three-dimensionalities on habitat preferences of organisms and enhances effective management in riverine ecology.

2 | STUDY REACH

The study area is the Ziller River, with a groyne field stretch including depositional bars. The river is a tributary of the Inn River with approximately 56 km length, located in the alpine Ziller valley in Tyrol, Austria (Figure 1). Four large reservoirs in the Ziller basin supply water to eight HPPs generating about 1,200 GWh/year. Based on the European Union Water Framework Directive (EU-WFD) classification the river is having a “poor ecological status” due to multiple anthropogenic impacts. In addition, as a result of existing barriers, the downstream river stretches, that is, Inn River, have serious bed incision problems due to the lack of sediment supply from upstream.



FIGURE 1 Location of study area, inset: Ziller valley and the corresponding waterway network [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ra.4035)]

Nevertheless, hydropeaking is a major considerable ecological problem in Ziller River with ramping ratios between base and peak flow of $>1:20$ in winter (Hauer, Unfer, Holzapfel, Haimann, & Habersack, 2014). Thus, a river reach near the village Ramsau (located at $47^{\circ}12'N$ $11^{\circ}52'$) is selected for the current investigations. Here, the HPP Mayrhofen located upstream regularly affects the Ramsau reach (average slope 0.0054) by sudden water level fluctuations due to hydropeaking (sometimes multiple times per day). As superimposition of this intense hydrological pressure, the Ziller River has undergone considerable morphological degradation due to flood protection measures in the 1960s, which left the fragile ecosystem more vulnerable to sudden flow alterations (cf. Hauer et al., 2014).

3 | METHODS

The applied methods are split into three main parts: (1) the macroinvertebrate habitat assessment as an important input reflecting the onsite abundance and biomass of two selected species, (2) the transformation of the field data into habitat suitability curves used for numerical habitat modelling, which is (3) based on the novel integrative assessment using a three-dimensional unsteady hydrodynamic model.

3.1 | Macroinvertebrate habitat assessment

As a part of larger macroinvertebrates habitat investigations in the Ziller valley, the study reach was assessed based on the multi-habitat sampling (MHS) method using a WFD compliant net. In addition, hydraulics-specific sampling was performed with a modified Surber sampler (box Sampler). The investigation site was sampled using 18 replicates in different distances from the channel banks. All samples were conducted during base-flow, and all benthic invertebrates per sample were counted and determined to the lowest possible level.

Moreover, abiotic parameters such as mean velocity at 40% of the depth ($v_{40\%}$) and near-bed velocity (v_{nb}) were measured using an electromagnetic flow meter, that is, Marsh-McBirney FLOMATE™ 2000; corresponding water depth in sampling points were also acquired; details of habitat sampling are provided in Leitner et al. (2017). In the current study the near-bed velocities were also used further to verify near bed simulated velocities out of three-dimensional unsteady simulations. The geometry of the investigation reach, that is, cross-sections, breakpoints, groyne-fields, was determined precisely (Hauer et al., 2014).

3.2 | Habitat suitability curves

In order to model the hydraulic impact of the flow on macroinvertebrates communities the abiotic parameters were measured and analysed for the selected taxa. In the current work, the data was acquired based on conventional acquisition guidelines. The outputs regarding habitat suitability for selected taxa is presented as habitat suitability curves for two indicative taxa in Figure 2, that is, the Ephemeroptera genus *Rhythrogena* and the Trichoptera species *Allogamus auricollis*. Both taxa are typical elements of alpine streams and rivers and well known to inhabit distinct hydraulic habitats: species of the genus *Rhythrogena* explicitly prefer fast flowing stony areas, while *A. auricollis* is restricted to slowly flowing or even stagnant riverine habitats (Schülting et al., 2016; Waringer, Graf, & Malicky, 2013). Therefore, these antagonistic taxa were chosen to describe the investigated effects of pulse releases.

The suitability curves prepared are based on data of hydrologically unaffected and affected investigation reaches of Ziller River. It is based on the theoretical maximum current velocities during hydropeaking to demonstrate the variation of habitat preferences (Holzapfel et al., 2017). Additionally, the suitability curves based on bed shear stress values provided further shear stress relevant observations.

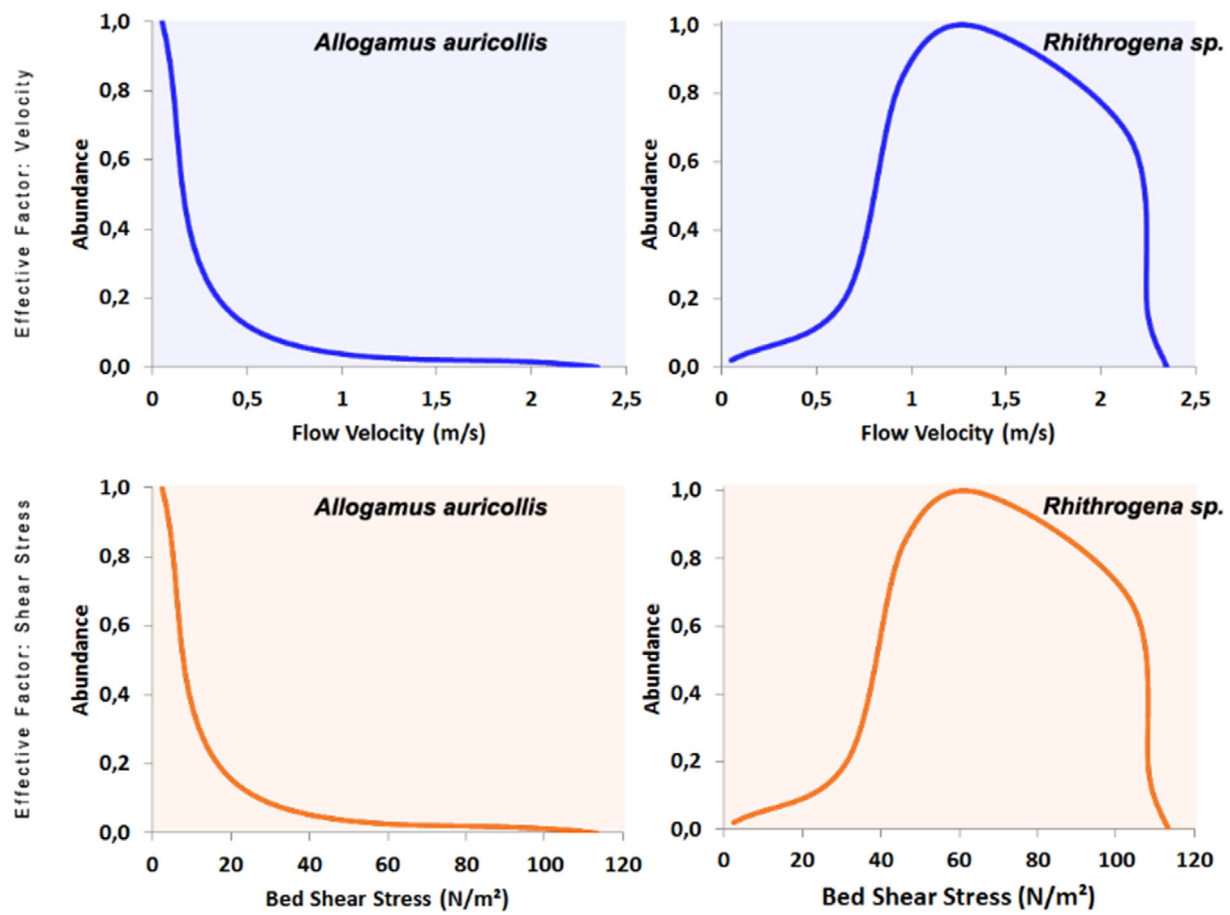


FIGURE 2 (above) Habitat suitability curves of *A. auricollis* and *Rhithrogena* sp. effective factor is the velocity (Holzapfel et al., 2017); (below) effective factor is the shear stress parameter calculated using Equation (2) [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jna.4035)]

Previously, the two-dimensional depth-averaged velocity approximation was applied as the modelling approach (Holzapfel et al., 2017; Leitner et al., 2017). Using 2D hydraulic models it is straightforward to apply the habitat suitability curves which are prepared based on mean depth-averaged velocity as effective factors. For three-dimensional CFD models, considering the time and effort required for such simulations, it is not practical to calculate the depth-averaged values based on the three-dimensional simulated available velocity vectors and lose a load of data in the averaging process. Therefore, in the current study a novel unsteady three-dimensional approach was applied to use another effective and representative factor for suitability curves to demonstrate the three-dimensionality of flow and provide more precision. Regarding the scope of this work, that is, macroinvertebrates, the bed shear stress found to be a good choice among the simulated data. Accordingly, the equation for bed shear stress is as follow (Statzner & Holm, 1989)

$$\tau_0 = \rho u_*^2, \quad (1)$$

where τ_0 is the bed shear stress (Nm^{-2}), u_* shear velocity (ms^{-1}), ρ density of water (kg m^{-3}). Shear velocity can be derived from the log-law velocity profile as follow,

$$u_* = 0.4 v_{40\%} (\ln y_{40\%} - \ln y_{nb})^{-1}, \quad (2)$$

$y_{40\%}$ and y_{nb} are distance of 40% depth of flow from the bottom and depth of the hydraulic wall region, respectively (Nezu et al., 1993). Here, bed shear stress is calculated using Equations (1), (2), and measured mean velocities from Leitner et al. (2017) and Holzapfel et al. (2017). The habitat suitability curves with bed shear stress as their effective factors are then generated based on the mentioned calculated shear stress.

3.3 | Application of three-dimensional unsteady numerical modelling

3.3.1 | Data pre-processing and model setup

The acquired data of the terrestrial survey (Hauer et al., 2014) and corresponding habitat data were already discussed and verified in previous studies (e. g. Leitner et al., 2017; Holzapfel et al., 2017). The high-resolution and verified topographic and bathymetric information were applied to generate high-quality digital elevation terrain model (DEM) using Ansys® Designmodeler™. The geometry was then exported to

generate the mesh system for further numerical simulations. The patch conforming algorithm for tetrahedrons was provided by the simulation software and applied to the current mesh. The algorithm ensures mesh refinement where it is necessary and allows faster computations. Considering the mesh convergence study, the study area was converted to a mesh structure of about 1.7 million tetrahedron elements (mesh resolution: max 99 cm to min 5 cm, the domain has imbalances of less than 1%); towards boundaries the mesh is refined down to 5 cm.

Hydraulic parameters of the reach were simulated using the finite volume approach. Model input parameters are summarized in Table 1.

A commercial three-dimensional CFD model, that is, Ansys®-CFX®, is used to solve the Reynolds-Averaged Navier–Stokes (RANS) equations and to perform a three-dimensional hydraulic simulation. The model solves both RANS equations for steady and unsteady states. RANS equations have Reynolds stress term, and accordingly there is a need for a Reynolds stress closure in the equation. Here, Reynolds Stress Model (RSM) was employed as Reynolds stress closure schemes for the RANS equations. The current CFD model consists of the boundary conditions discretized by a finite volume mesh. The boundary conditions include boundaries at the riverbed, banks, other

TABLE 1 Model input parameters for the study reach

$\frac{Q_{\text{mean}}}{(\text{m}^3 \text{ s}^{-1})}$	$\frac{Q_{\text{base}}}{(\text{m}^3 \text{ s}^{-1})}$	$\frac{Q_{\text{peak}}}{(\text{m}^3 \text{ s}^{-1})}$	$\frac{H}{(\text{m})}$	$\frac{h}{(\text{m})}$	$\frac{L}{(\text{m})}$	$\frac{I}{(\text{m/m})}$	$\frac{\nu}{(\text{m}^2 \text{ s}^{-1})}$	Fr	Re
29.30	3.9	93.90	3.4	0.65	281.9	0.0054	1.43×10^{-6}	0.54	4.13×10^5

Abbreviations: Fr , Froude number; H , bankfull depth; h , mean flow depth; I , channel slope; L , reach length; Q , Discharge; ν , kinematic viscosity; Re , Reynolds number.

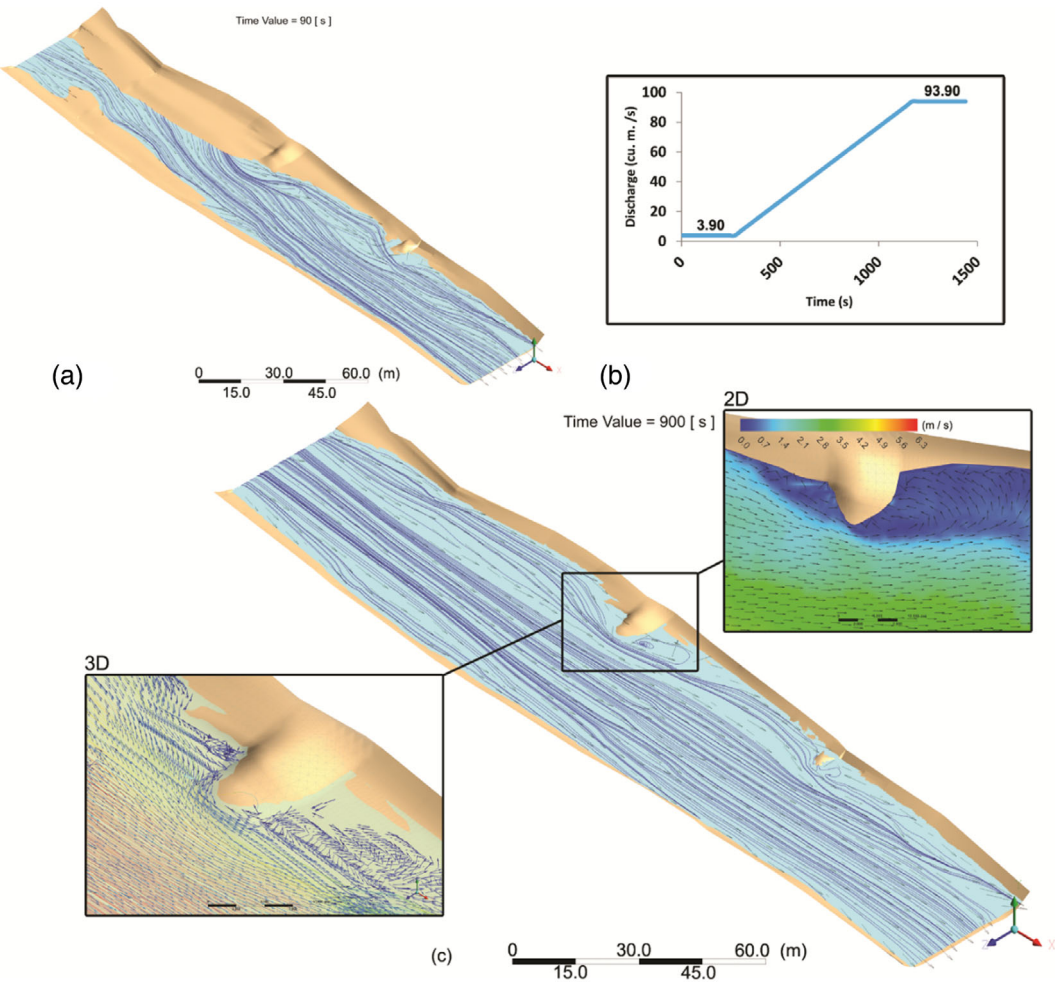


FIGURE 3 Flow simulation streamlines of the investigated reach (a) base flow condition 3.90 m³/s (b) unsteady flow hydrograph: in 900 seconds flow reaches the peak flow 93.90 m³/s (c) peak flow state: insets show distributions of vector fields in the proximity of the groyne filed as in 2D and 3D simulations [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

flow obstructions, as well as two open boundaries at the upstream and downstream of the modelled reach. The solver calculates free surface variations. Non-slip condition was satisfied at any solid boundary, that is, the flow velocity at the solid boundary was equal to the velocity of solid boundary. Additionally, the non-equilibrium law of the wall, which was expanded for rough walls and flow separations used to model the near-wall flow velocity field. Shear velocity was characterized by the turbulence intensity and wall shear stress was assumed to be constant within the inertial sublayer. The upstream boundary assigned to be unsteady based on a stage-discharge curve for simulations. In addition, at the upstream, a hydrostatic pressure profile was established up to the water surface elevation. A hydrostatic pressure profile, which was consistent with the known downstream water depth, based on stage-discharge from the downstream cross-section was applied as the downstream boundary.

3.3.2 | Modelling boundaries and parametrization of the analysis

The primary simulation was performed for the selected reach at $3.90 \text{ m}^3\text{s}^{-1}$ [Mean annual low flow] and the corresponding isometric view is demonstrated in Figure 3a. To simulate a hydropeaking event, a rising limb of the corresponding hydrograph was implemented into the model (Figure 3b). The isometric depiction of three-dimensional unsteady simulation results at peak-flow, that is, $93.90 \text{ m}^3\text{s}^{-1}$ is provided in Figure 3c; two insets demonstrate availability of velocity vectors information in 2D and three-dimensional views. The increase from $3.90 \text{ m}^3\text{s}^{-1}$ up to $93.30 \text{ m}^3\text{s}^{-1}$ is the worst-case scenario in terms of hydropeaking and only occurs during winter low-flow conditions. For the analysis, the applied three-dimensional simulations

provide velocity vectors, including velocity direction and magnitude, and their distribution along the depth of the flow. Considering the availability of three-dimensional unsteady flow velocity and all the turbulence tensors, it is quite easy to depict vertical and horizontal vortex structures, calculated Reynold's shear stress, and solid boundary shear stresses. The mentioned parameters are partially or totally unavailable in 2D simulations.

Three-dimensional simulations generate extensive and detailed results. To provide a better insight into the results and to ensure analytical accuracy for the interpretation of benthic drift, a number of selected points in the geometry were chosen to represent the simulation Results. The investigation points were selected based on the following procedure: (a) habitat maps based on 2D modelling results (Holzapfel et al., 2017) were used to generally describe the potential habitat use; (b) investigation points were selected randomly but with the condition of being in high suitable areas ($\text{HSI} > 90\%$) shown in habitat maps (Figure 4). The investigation points depict habitat diversity of the river reach, that is, banks, river bed, and proximity of groyne fields (see Figure 5). Further detailed hydraulic analyses were performed for these specific points.

Schülting, Feld, Zeiringer, Hudek, and Graf (2019) implied the tendency of macroinvertebrates to enter the flow before the maximum peak. In other words, there is an undescribed behaviour for some macroinvertebrate taxa to join the flow and drift during the rising limb before the maximum flow. It is assumed that there are certain physical parameters, which trigger the drift of macroinvertebrates. To address this phenomenon, the simulated values were extracted from the "observation point" (Figure 5). Using the derived data, instantaneous three-dimensional velocity components, that is, streamwise (u), lateral (v) and vertical (z) velocities, at the corresponding points (5 cm above the riverbed) were provided.

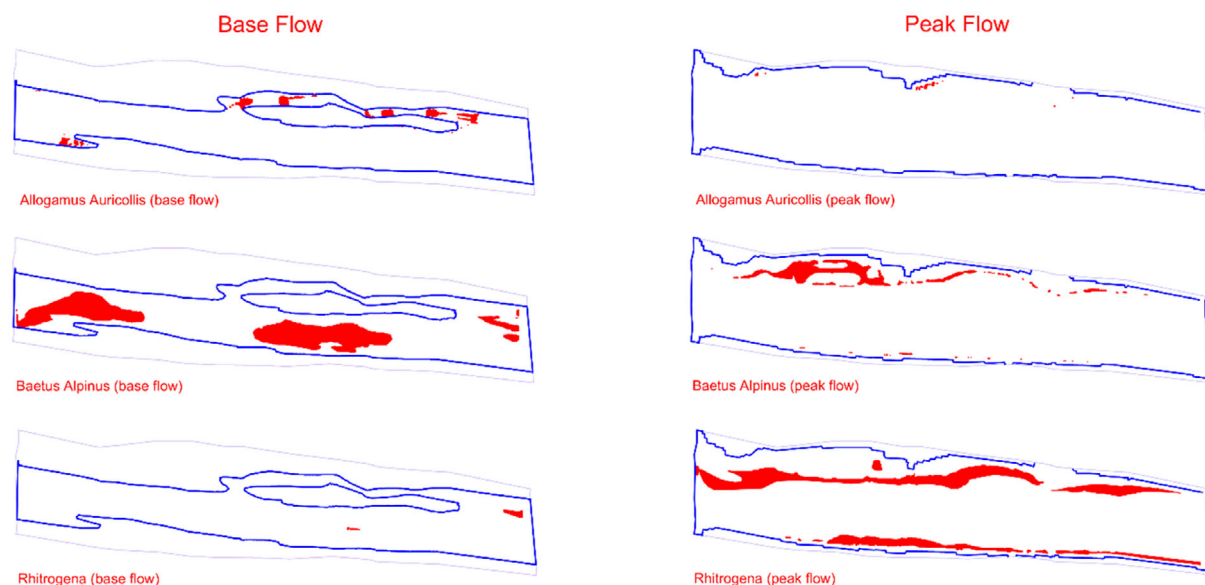


FIGURE 4 Red patches show areas with $\text{HSI} > 90\%$ from 2D habitat model for the studied species at peak and base flow. Wetted area is highlighted with blue lines [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jra.4035)]

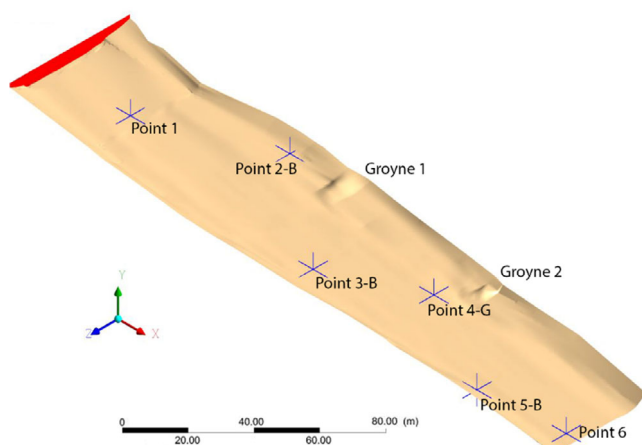


FIGURE 5 selected observation points and their location in three-dimensional simulations geometry [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jra.4035)]

4 | RESULTS

4.1 | Habitat suitability using three-dimensional unsteady simulation

In the following based on the numerically simulated shear stress and the transferred suitability curves of Leitner et al. (2017) and Holzapfel et al. (2017) the habitat suitability for *A. auricollis* and *Rhithrogena* sp. was evaluated in Ziller River. Based on the three-dimensional unsteady hydrodynamic simulation of low and high flow due to hydropeaking in the investigation reach the habitat suitability maps are provided. The maps discern the changing habitat suitability of a river reach in the course of hydropeaking time due to shear stress alterations.

4.1.1 | Modelling results for *A. auricollis*

The alteration from base to peak flow in the Ziller River happens within 15 minutes (Hauer et al., 2014). HSI values for *Allagamus auricollis* in the base flow condition, that is, 90 s, and the peak flow at 900 s are presented in Figure 6. As expected, by increasing flow discharge and accordingly velocity and depth the suitable habitat for the mentioned species remains along bankside. However, as it can be discerned in Figure 6 the small refuge patches due to slight changes in the reach topography are provided by the unsteady three-dimensional simulations. The provided suitability maps include complete flow column data, and their instantaneous alterations during the hydropeaking curve. In contrast the two-dimensional depth-average models do not provide such information.

4.1.2 | Modelling results for the genus *Rhithrogena*

The simulation results were applied to provide a detailed habitat suitability map for *Rhithrogena* sp. (Figure 7). As expected, the suitably

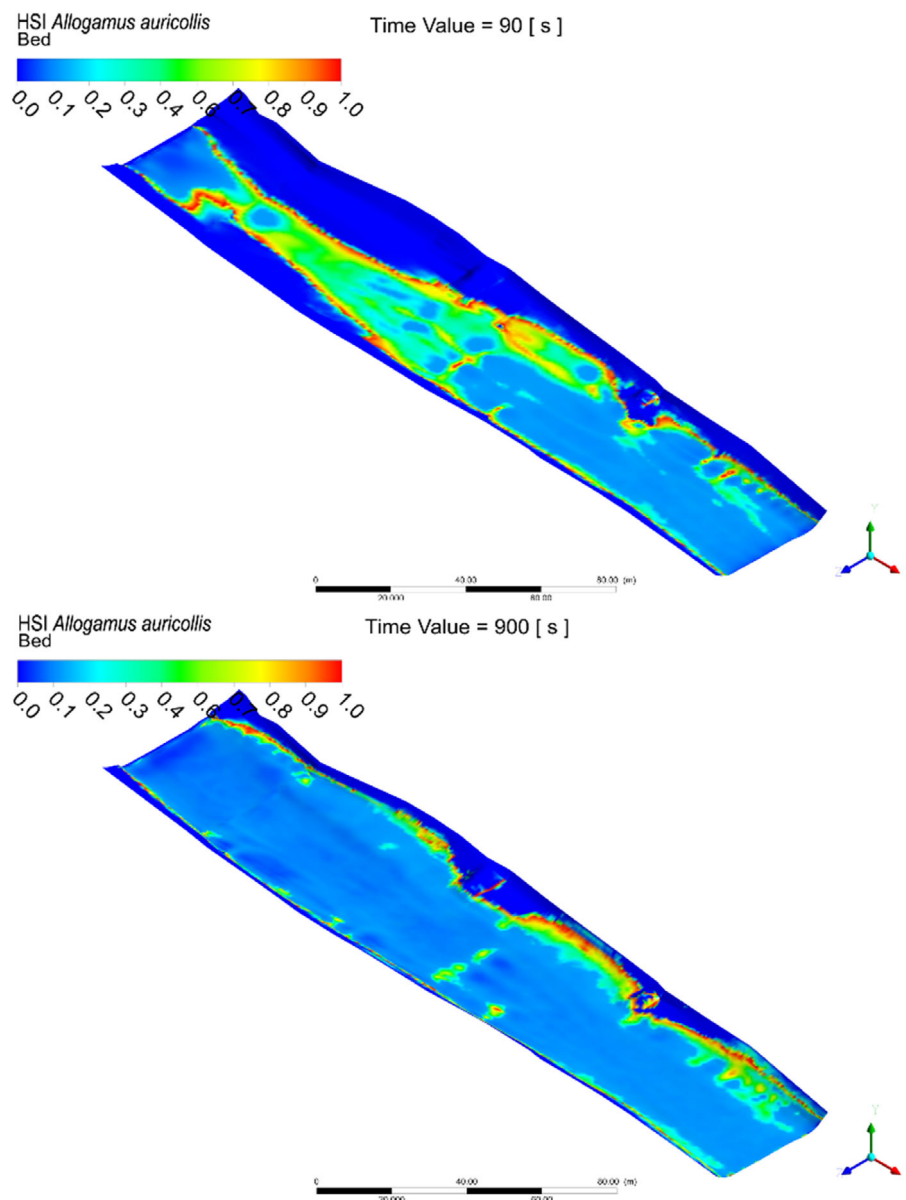
index discerns an opposite pattern in comparison to *A. auricollis*, which is a stagnophilic species. The results for *Rhithrogena* sp. reveal the high flow-tolerance of this rheophilic taxon. Obviously, this outcome is not involving abundance and biomass parameters, although a significant decrease of individuals is obvious in field investigations (Leitner et al., 2017). In addition to the ecological interpretations of the results, one can discern considerable resolution of the generated map based on unsteady flow and shear stress simulations (see videos provided in the Appendix); the details provided here are usually ignored in the case of classical habitat mapping using conventional one- or two-dimensional models.

4.2 | Analysis of unsteady aspects in parameter changes

Figure 8 provides streamwise and vertical components of velocity for the “observation point”. Due to characteristics of natural rivers, at the first glance, the selected observation points are showing a large range of asymmetry. Therefore, to clarify the outcome, a location-classification for them is necessary. Points 1 and 6 are located at the inlet and outlet of the model near to deepest parts of the river. Other points are located rather near to the banks (B), however, points no. 2 and 4 are also affected by extreme vorticities before and after the groyne field (G).

The streamwise velocity component (u) shows the creation and propagation of a wave front due to hydropeaking. The peak of the streamwise velocity corresponds to the time, in which the wave front reaches the certain “observation point”. However, the location of observation point, that is, B, G, and BG, should be considered. The streamwise velocity generally contributes to more than 90% of the velocity magnitude (scalar absolute value of velocity components) in straight channels (Blanckaert & Graf, 2001). In case of increased flow three-dimensionality, which are quite common in natural streams, for example, flow around natural obstructions or in the proximity of boundaries, the contribution percentage of velocity components change drastically (Farhadi, Sindelar, et al., 2018). This phenomenon is also discernible in Figure 8; streamwise velocity at points 1 and 6 are from a higher order of magnitude in comparison to vertical velocity components. However, this pattern changes toward banks (B) and the Groyne field (G). The peaks of the streamwise flow are followed by a slight decrease and then an almost constant value toward the end of the simulation, which is obvious at point 1. Streamwise velocity components peak time in the Figure 8 corresponds to the time of arrival of hydropeaking wavefront to these points. This observation will be explained in detail further in discussion. The peak reaches to the point 6 a few hundred seconds later. Point 2 is affected by the groyne field; Figure 3c shows the deviation of the streamline from the groyne field to the mainstream, which will affect the streamwise velocity drastically. However, the peaks of the streamwise velocity confirm the arrival of the wavefront to the locus of point 2. Point 3 and 5 are also affected by the hydraulic bank processes. Nevertheless, the peak of the streamwise velocity can be observed correspondingly.

FIGURE 6 Alterations of HSI values for *Allogamus auricollis* calculated based on three-dimensional simulations from approximately base flow (above) to the hydropeaking maximum flow (below) [Color figure can be viewed at wileyonlinelibrary.com]



Also, the vertical velocity component demonstrates a trend. The minimum vertical velocity is occurring about two simulation-step before the maximum streamwise velocity. Near the boundaries and in the groyne field, the vertical velocity is higher and contributes more to the sum of velocity magnitude. At points 1 and 6, along the main-stream, also streamwise and vertical velocity parameters show a reverse trend, that is, increasing streamwise velocity and a decrease of the vertical component are happening consequently.

Based on the simulations, shear stress fluctuations at the riverbed were derived using Equation (1). In the study reach the depth of wall region y_{nb} is calculated to be about 5×10^{-4} to 3×10^{-3} m. Velocity distribution in the wall region is linear; further to the surface and beyond the wall region, aka “the outer region”, velocity distribution is generally assumed to be logarithmic (see Equation 2). The size of an *A. auricollis* larva is reported to be about 2×10^{-3} m (Waringer et al., 2013) and a larva of *Rhithrogena* sp. is about 1.5×10^{-3} m

(Vidinova, 2003). Therefore, a considerable population of the investigated taxa is either fully or at least have a large part of their body mass submerged in the hydraulic wall region of the channel, where the velocity distribution is linear.

The derived riverbed shear stress values are shown in Figure 9, the time span begins at the peak value for point 1, that is, 220 s. For every observation point, shear stress values are rather in line with wave front propagation. They show a peak of shear stress which is either decreased in time to a more stable value or slightly increased and stabilised after a few more time steps. The wall shear stress values and fluctuations are decisive parameters in fine sediment transport and consequently has an undoubtable effect on catastrophic drift of macroinvertebrates (see Waringer, 1993; Bacher & Waringer, 1996; Möbes-Hansen & Waringer, 1998). For the points 2, 3, 4, and 5 the shear stress fluctuation is more discernible, which is related to groyne field three-dimensionality. Point 6 shows only the

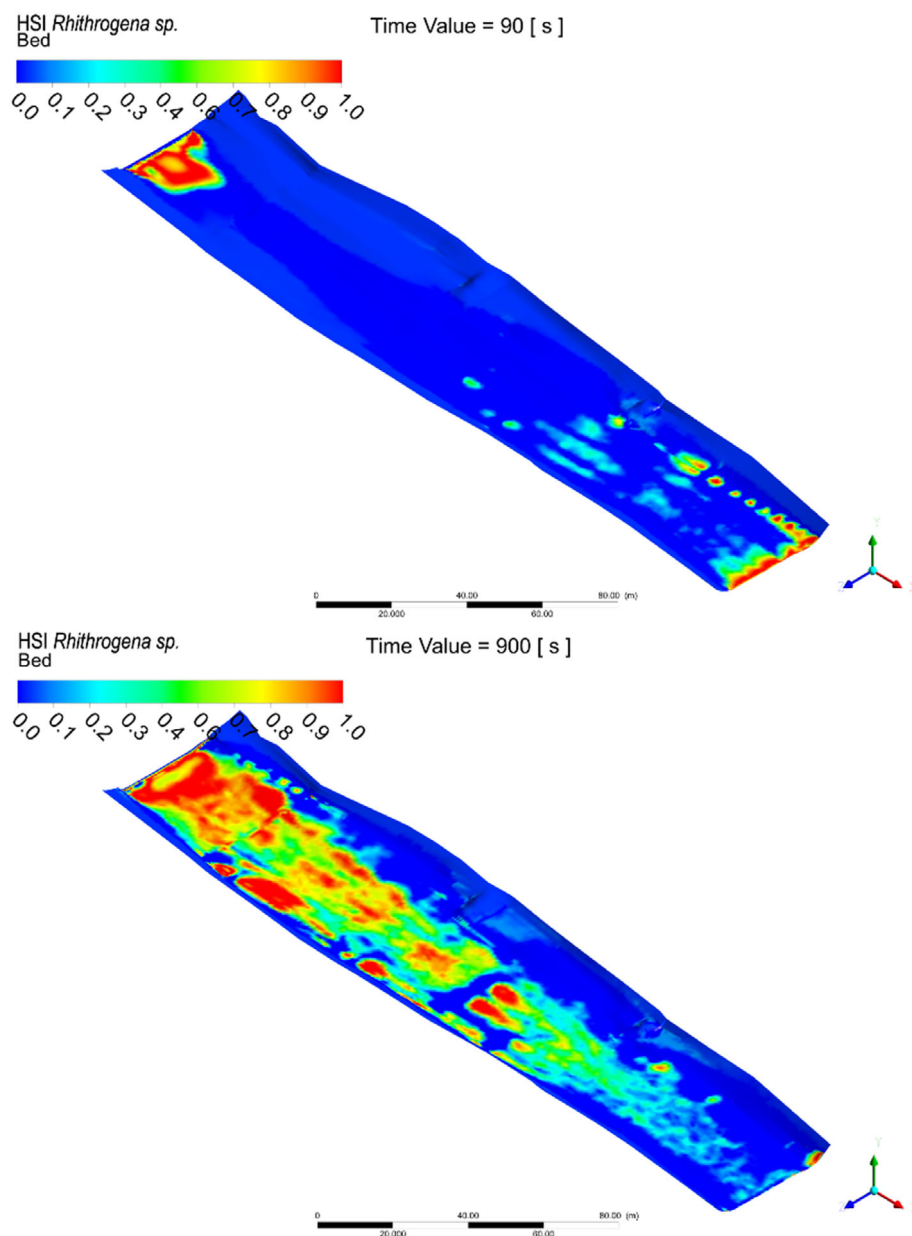


FIGURE 7 Alterations of HSI values for *Rhithrogena* calculated based on three-dimensional simulations from approximately base flow (above) to the hydropeaking maximum flow (below) [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jra.4035)]

first peak and due to restriction of simulation time further possible fluctuations are not simulated for this specific point.

5 | DISCUSSION

The scarcity of three-dimensional model applications in habitat simulations caused potential impacts of abiotic parameters on habitat properties to be left unregarded. Two-Dimensional simulations are widely used and shown to be reliable, however, it ignores flow three-dimensionality and consequently simplifies habitat procedures (Pisaturo et al., 2017). As shown, the amount of information that will be lost due to the simplification of 2D simulations is noticeable.

The current study does not analyse the macroinvertebrates drift processes however the outcomes can provide valuable foresight for

further studies. Catastrophic drift, which removes many individuals from the riverbed and entrain them into the water column can be related to anthropogenic activities such as hydropower production and corresponding hydropeaking events (Bruno, Maiolini, Carolli, & Silveri, 2010). It is a very likely scenario for the current study in the investigated reach of Ziller River, where within a few minutes HSI values alter drastically. In addition to HSI, the alterations of shear stress values are also provided in our study. It is known that the turbulent and corresponding flow instantaneous fluctuations have a considerable effect on macroinvertebrates drift (Hart & Russ, 1996), however due to the intrinsic complexity of the turbulent flow, a reliable perception of the effects of flow fluctuations on drift phenomena is not available. Based on high-resolution three-dimensional unsteady velocity measurements (Blanckaert, Garcia, Ricardo, Chen, & Pusch, 2013), it was shown that macroinvertebrates are not only

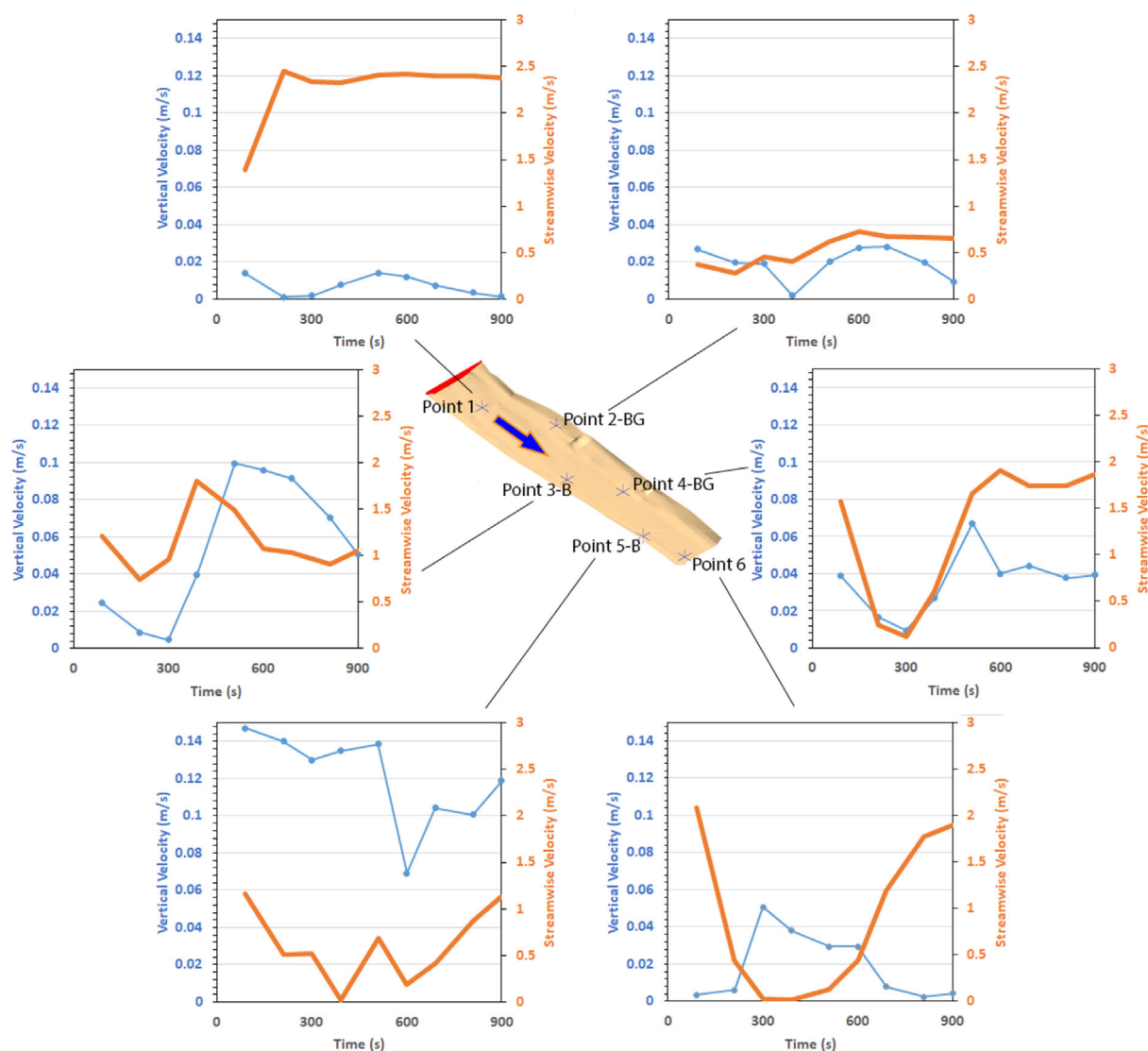


FIGURE 8 simulated vertical (left axis) and streamwise (right) velocity components for the simulated points [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

affected by the flow peak values but also instantaneous flow fluctuations play a significant role in drift relevant processes. In the mentioned study, the shear stress at the time of entry was reported to be about 300% higher than the time-average value. The current work shows the reliability of three-dimensional unsteady simulations to address temporal alterations of flow and shear stress (see Figure 9). The areas with expected high flow three-dimensionality, that is, in the proximity of the banks (point 3B) and near to the groyne structures (point 4BG), demonstrate very high shear stress alterations before reaching the peak flow. 2D steady or quasi-unsteady models cannot demonstrate mentioned flow complexities due to their computational characteristics.

It is observed that the simulated streamwise velocities reach their maximum values earlier than the water depth peak. Accordingly, Song and Graf (1996) observed this time lag between velocity and water depth peaks in an experimental flume; Shen and Diplas (2010) found a

similar streamwise velocity trend based on an investigation and discussed that these observations are more pronounced in riffles of a natural stream. In the current observation the phenomenon is more perceptible for the points near to the banks and groyne field with lower water depths and higher flow three-dimensionality, that is, 3B & 4BG. In the mentioned shallow points and during the peak wave, velocities are higher and water surface slopes are steeper; after passage of the peak wave, the water surface slope and the streamwise velocity decrease. However, deeper parts of the stream with larger volumes can compensate the change and handle the impact of the wave with smaller changes in the water surface slopes. In addition, effects of the flow three-dimensionality, that is, secondary flows, turbulence intensities, etc., are needed to be considered in the current observation.

In the current study, it is shown that another simulated hydraulic indicator, the vertical velocity component, has considerable hydraulic

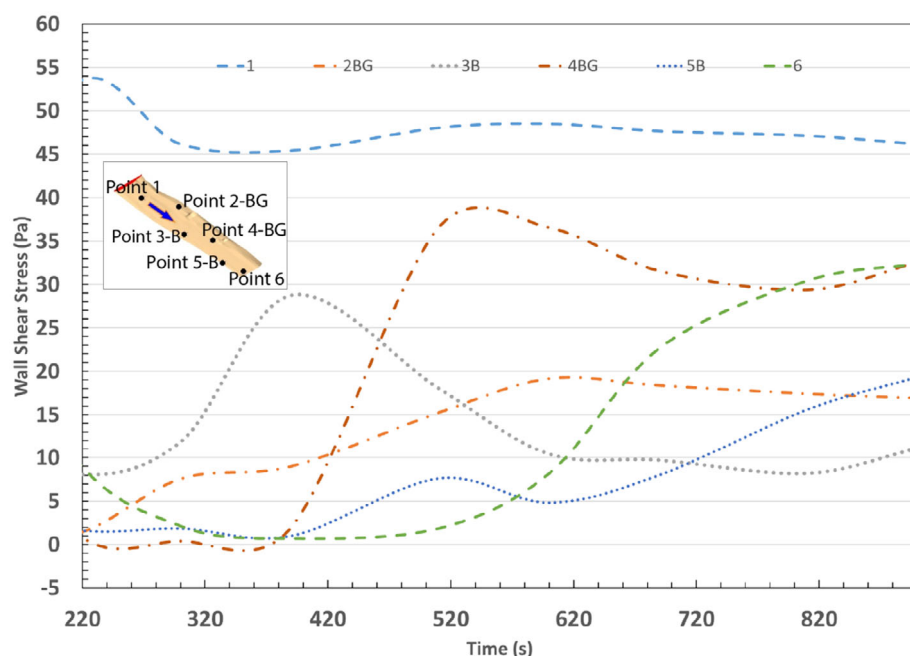


FIGURE 9 Bed shear stress at “observation point”; inset: the location of the points in investigation area are demonstrate. Labels “B” and “G” shows the proximity of points to the Bank and Groyne respectively [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jra.4035)]

effects and should definitively be considered in habitat investigations. The simulated hydraulic properties can be considered in studies addressing the dislodgement of some macroinvertebrate's species into the water column before or during the peak of the hydropeaking event. The phenomenon is currently under investigation and considered to be an open question in macroinvertebrates investigations.

6 | CONCLUSIONS

In comparison to the habitat suitability investigations using two-dimensional quasi-steady numerical simulations, the current unsteady simulations have two apparent advantages:

- instead of simulating the increasing flow by constant interrupted sequential steps, the current simulations calculate flow properties during the transition time of an unsteady increasing flow rate. Therefore, the increased energy slope due to the constant increase of flow discharge is considered.
- The three-dimensional simulations provide a decisive factor in habitat investigations, that is, vertical velocity component, which cannot be addressed using two-dimensional simulations. Vertical velocity component is critical in macroinvertebrate drifts due to flow alterations (cf. Statzner & Holm, 1989). Availability of all components of the flow velocity, also made wall shear stress calculations possible.

The mentioned results and discussions suffer from two major shortcomings.

- The current results were not based on a fully calibrated simulation. However, the outcomes which are discussed here are not affected directly by the fine tunings in a calibration process. This is not the

case for the two-dimensional models which were used in this study for comparison. They were calibrated according to the discharge/water stage relationship and as far as possible it was used in the current study.

- The current habitat indices were acquired for the common habitat models and did not consider possible flow-related alterations of habitats. Therefore, if not impossible, it is very difficult to come to a concrete conclusion based on the findings of this opinion paper.

On the other hand, our approach could provide an initiation for further investigations. The mentioned shortcomings will not hinder the conclusion that three-dimensional simulations will provide essential information for habitat modelling and can address open questions regarding habitat modelling and macroinvertebrates behaviours during hydropeaking.

DATA AVAILABILITY STATEMENT

The data that support the findings is available in the repository of the Institute of Hydrobiology and Aquatic Ecosystem Management, BOKU University, Vienna. The data is available upon request, however, there is no public URL available.

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APPENDIX A

1. Unsteady Simulation—Water Surface—Ziller River—Tyrol/Austria. <https://www.youtube.com/watch?v=ZCuroFLMHos>
2. Unsteady Simulation—Bed shear stress—Ziller River, Tyrol/Austria. <https://www.youtube.com/watch?v=fMrJ2St5Tp8>